CYCLE 1 PROGRAMS

PROGRAM 1: Simulation and display

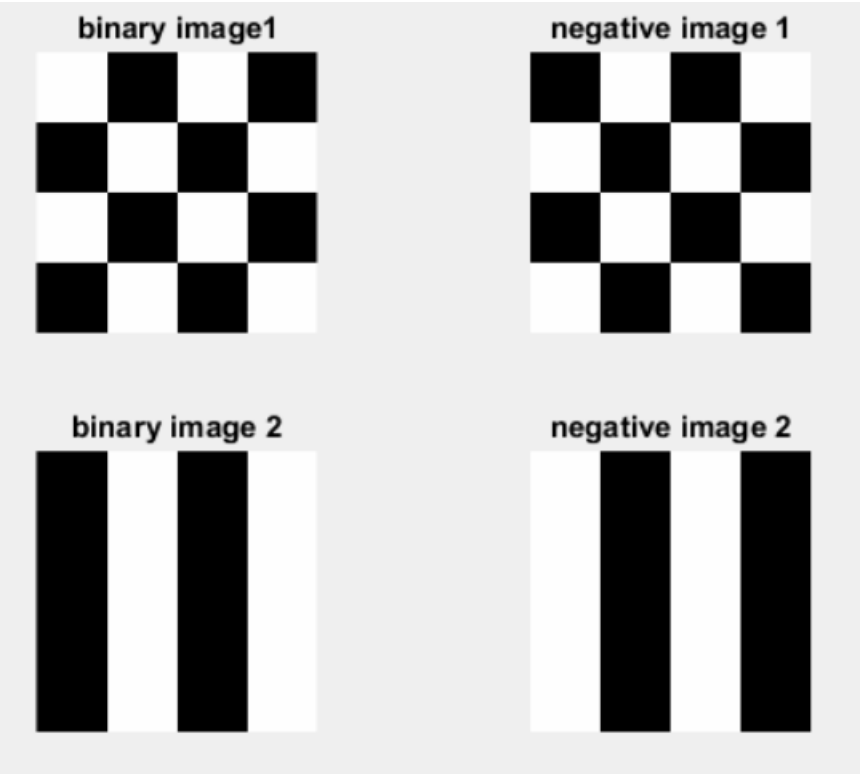
1.1 Simulation and display of Binary image and its negative

CODE:

```
w=ones(64,64);
b=zeros(64,64);
i=[w,b,w,b;b,w,b,w;w,b,w,b;b,w,b,w];
subplot(2,2,1);
imshow(i);
title('binary image1');
subplot(2,2,2);
imshow(~i);
title('negative image 1');
subplot(2,2,3);
w=zeros(64,64);
b=ones(64,64);
z=[w,b,w,b;w,b,w,b;w,b,w,b;w,b,w,b];
imshow(z);
n=size(z)
title('binar
y image 2');
subplot(2,2,4);
imshow(~z);
```

title('negative image 2');

OUTPUT:

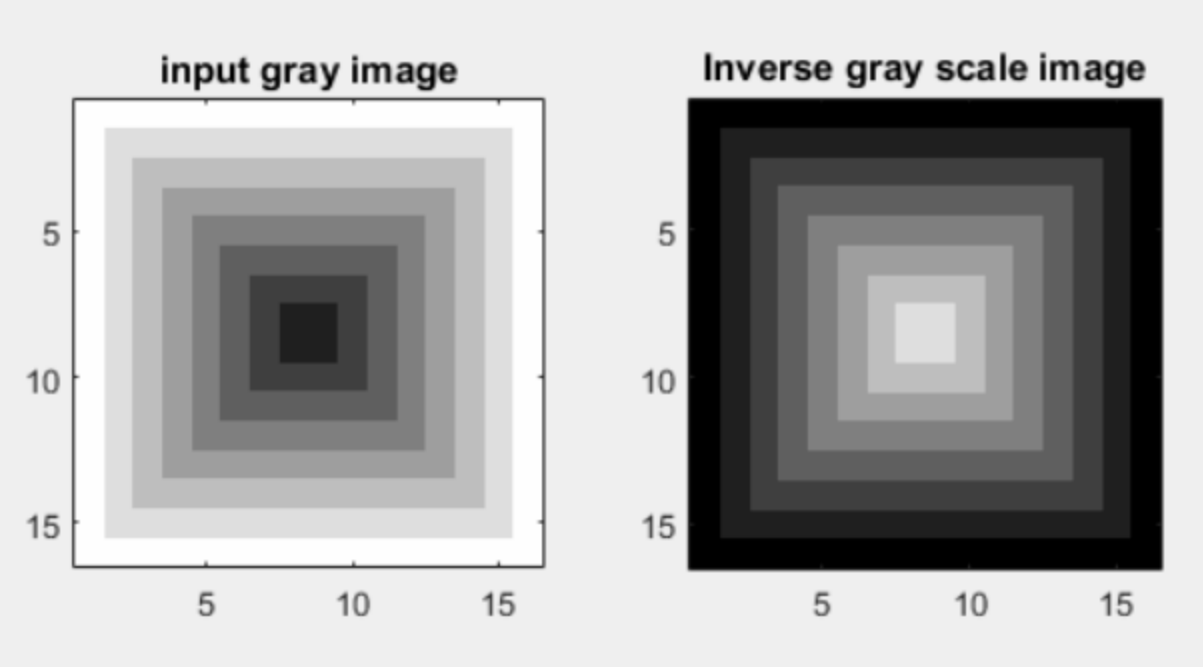


1.2. Simulation and display of gray scale image and its negative

CODE:

```
clc;
clear all;
a=[16 16 16 16 16 16 16 16 16 16 16 16 16 16 16 16];
b=[16 14 14 14 14 14 14 14 14 14 14 14 14 14 14 16];
c=[16 14 12 12 12 12 12 12 12 12 12 12 12 12 14 16];
d=[16 14 12 10 10 10 10 10 10 10 10 10 10 12 14 16];
e=[16 14 12 10 8 8 8 8 8 8 8 10 12 14 16];
f=[16 14 12 10 8 6 6 6 6 6 6 8 10 12 14 16];
g=[16 14 12 10 8 6 4 4 4 4 6 8 10 12 14 16];
h=[16 14 12 10 8 6 4 2 2 4 6 8 10 12 14 16];
X=[a;b;c;d;e;f;g;h;h;g;f;e;d;c;b;a];
I=X/16;
imwrite(I,'gray image.tif');
I=imread('gray image.tif');
NI=imadjust(I,[0,1],[1,0]);
figure
subplot(1,2,1);
subimage(I);
title('input gray image');
subplot(1,2,2);
subimage(NI);
title('Inverse gray scale image');
```

OUTPUT:


input gray image
Inverse gray scale image
5
10
15
5
10
15
5
10
15
5
10
15

PROGRAM 2: Implementation of relationships between the pixels

2.1 Arithmetic & logical operations on binary images

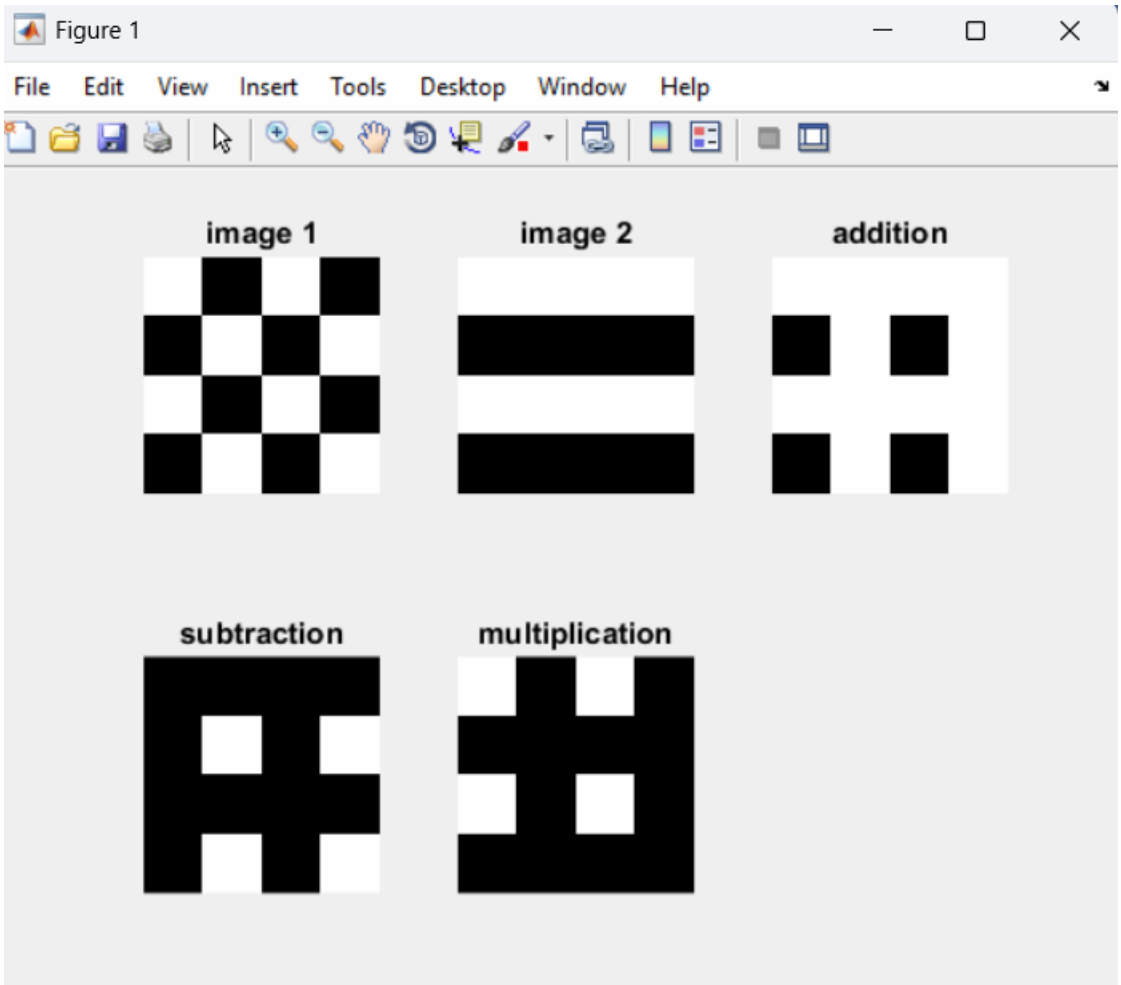
CODE:

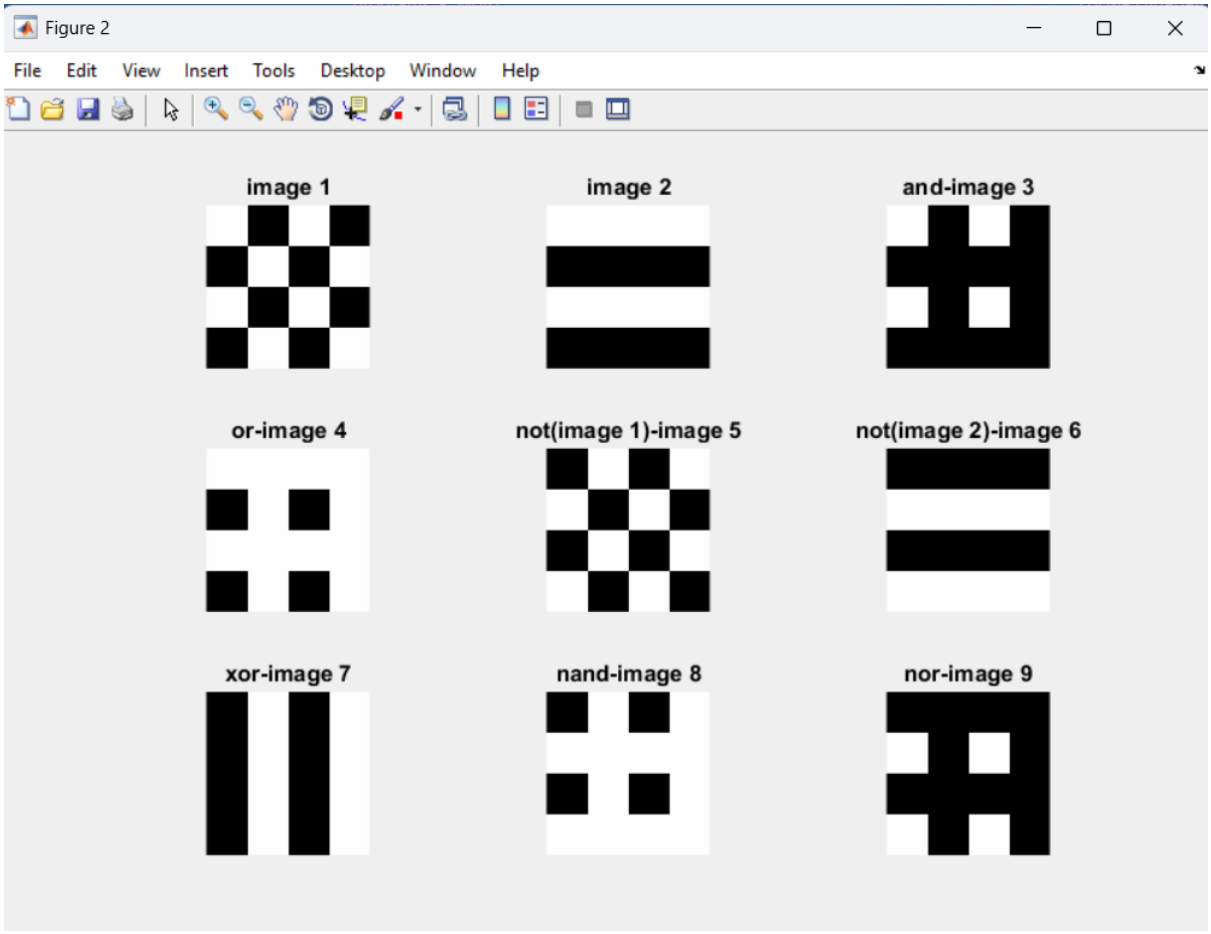
```
close all;
clear all;clc;
 i1=imread('C:\Program Files\MATLAB\MATLAB Production
Server\R2015a\bin\MIPIMAGES\bin_image1.tif');
subplot(2,3,1);imshow(i1);title('image 1');
i2=imread('C:\Program Files\MATLAB\MATLAB Production
Server\R2015a\bin\MIPIMAGES\bin_image2.tif');
subplot(2,3,2);imshow(i2);title('image 2');
i10=i1+i2;
subplot(2,3,3);imshow(i10);title('addition');
i11=i1-i2;
subplot(2,3,4);imshow(i11);title('subtraction');
i12=i1.*i2;
subplot(2,3,5);imshow(i12);title('multiplication');

figure(2);
subplot(3,3,1);imshow(i1);title('image 1');
subplot(3,3,2);imshow(i2);title('image 2');
i3=and(i1,i2);
subplot(3,3,3);imshow(i3);title('and-image 3');
i4=or(i1,i2);
subplot(3,3,4);imshow(i4);title('or-image 4');
i5=not(i1);
subplot(3,3,5);imshow(i5);title('not(image 1)-image 5');
i6=not(i2);
subplot(3,3,6);imshow(i6);title('not(image 2)-image 6');
i7=xor(i1,i2);
subplot(3,3,7);imshow(i7);title('xor-image 7');
i8=not(i3);
```

```
subplot(3,3,8);imshow(i8);title('nand-image 8');
i9=not(i4);
subplot(3,3,9);imshow(i9);title('nor-image 9');
```

OUTPUT:




Figure 2
File Edit View Insert Tools Desktop Window Help
image 1
image 2
and-image 3
or-image 4
not(image 1)-image 5
not(image 2)-image 6
xor-image 7
nand-image 8
nor-image 9

2.2 Neighbors of a pixel

CODE:

```
close all;
clear all;clc;
a=magic(5),a;
b=input('enter the row number');
c=input('enter the column number');
N4=[a(b,c-1),a(b-1,c),a(b,c+1),a(b+1,c),],N4;
Nd=[a(b-1,c-1),a(b-1,c+1),a(b+1,c+1),a(b+1,c-1)],Nd;
N8=[N4,Nd],N8;
```

OUTPUT:

```
a =
    17    24     1     8    15
    23     5     7    14    16
     4     6    13    20    22
    10    12    19    21     3
    11    18    25     2     9

enter the row number3
enter the column number3

N4 =
     6     7    20    19


Nd =
     5    14    21    12


N8 =
     6     7    20    19     5    14    21    12
```

2.3 Distance measures

CODE:

```
close all;
clear all;clc;
i=imshow('cameraman.tif');
impixelinfo;
[x,y]=ginput(2)
Ed=[(x(2)-x(1))^2+(y(2)-y(1))^2]^0.5;
disp('Euclidean distance=');disp(Ed);
D4=[abs(x(2)-x(1))+abs(y(2)-y(1))];
disp('City block distance=');disp(D4);
D8=max(abs(x(2)-x(1)),abs(y(2)-y(1)));
disp('Chess board distance=');disp(D8);
```

OUTPUT:



Pixel info: (X, Y) Intensity

```
x =
   44.0000
  217.0000
```

```
y =
     52
    195

Euclidean distance=
  224.4504

City block distance=
    316

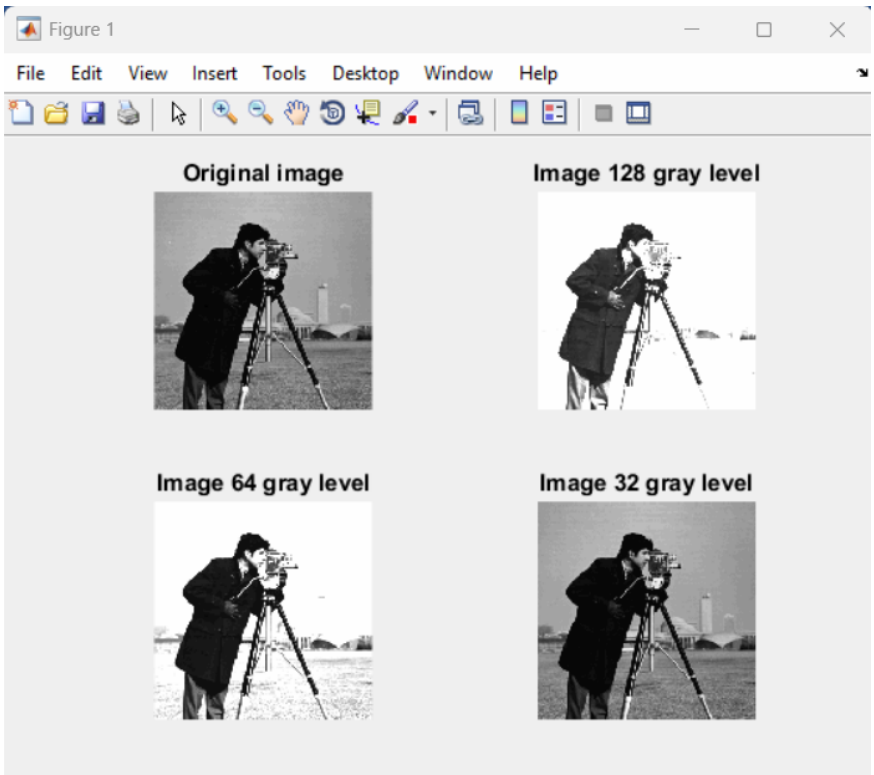
Chess board distance=
    173
```

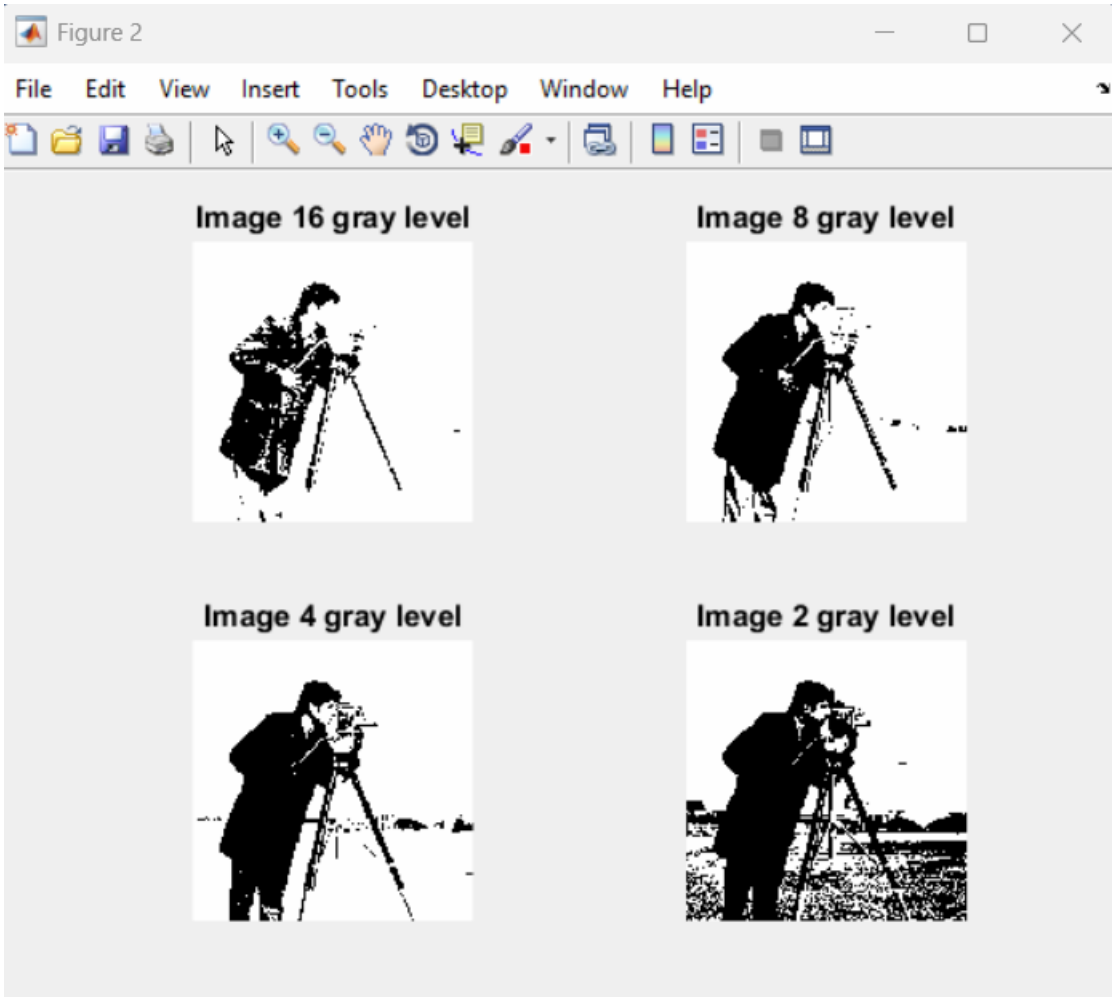
2.4 False contouring effect

CODE:

```
close all;
clear all;clc;
i=imread('cameraman.tif');
figure(1);
subplot(2,2,1);imshow(i);title('Original image');
subplot(2,2,2);imshow(grayslice(i,128),gray(128));title('Image 128 gray
level');
subplot(2,2,3);imshow(grayslice(i,64),gray(64));title('Image 64 gray level');
subplot(2,2,4);imshow(grayslice(i,32),gray(32));title('Image 32 gray level');
figure(2);
subplot(2,2,1);imshow(grayslice(i,16),gray(16));title('Image 16 gray level');
subplot(2,2,2);imshow(grayslice(i,8),gray(8));title('Image 8 gray level');
subplot(2,2,3);imshow(grayslice(i,4),gray(4));title('Image 4 gray level');
subplot(2,2,4);imshow(grayslice(i,2),gray(2));title('Image 2 gray level');
```

OUTPUT:

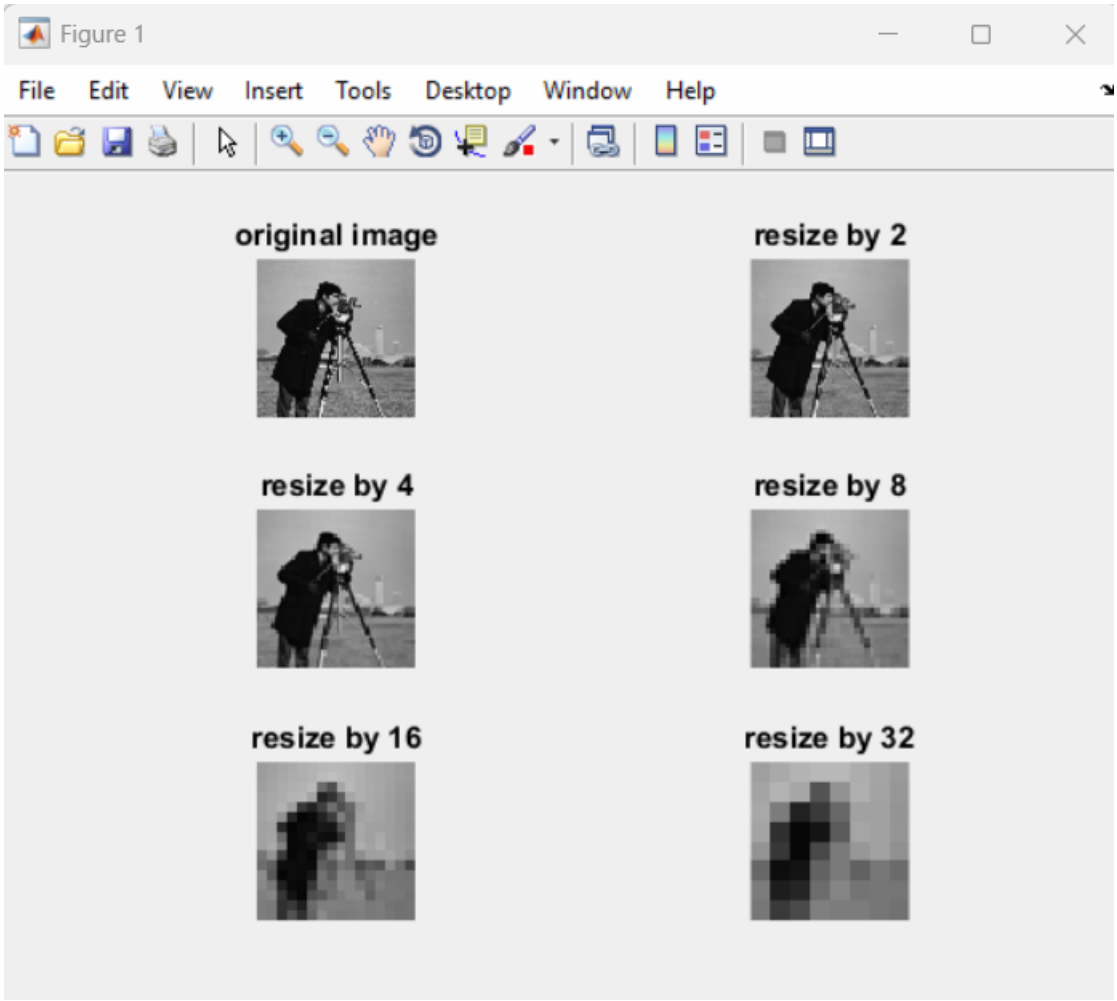



Figure 2
File Edit View Insert Tools Desktop Window Help
Image 16 gray level
Image 8 gray level
Image 4 gray level
Image 2 gray level

2.5 Checkerboard effect

```
clear all;
close all;clc;
X=imread('cameraman.tif');
subplot(3,2,1);
imshow(X);title('original image');
subplot(3,2,2);
imshow(imresize(X,1/2));
title('resize by 2');
subplot(3,2,3);
imshow(imresize(X,1/4));
title('resize by 4');
subplot(3,2,4);
imshow(imresize(X,1/8));
title('resize by 8');
subplot(3,2,5);
imshow(imresize(X,1/16));
title('resize by 16');
subplot(3,2,6);
imshow(imresize(X,1/32));
title('resize by 32');
```

OUTPUT:

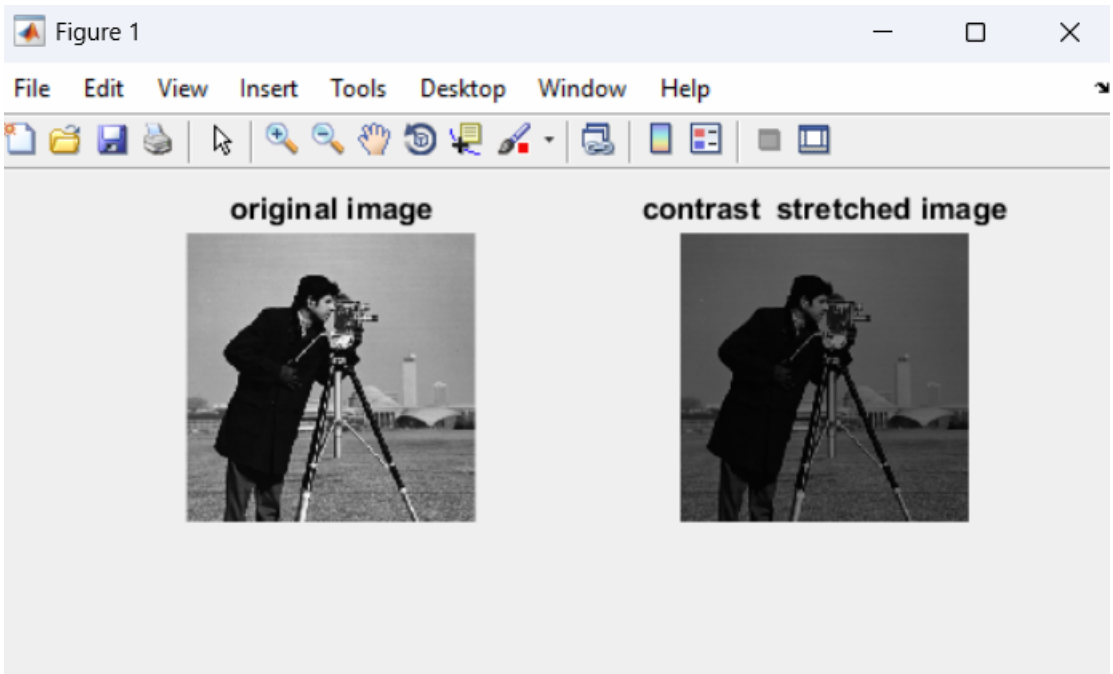

Figure 1
File Edit View Insert Tools Desktop Window Help
original image
resize by 2
resize by 4
resize by 8
resize by 16
resize by 32

PROGRAM 3: Contrast stretching, Intensity slicing, power law transformation

3.1 Contrast stretching

```
clear all;
close all;
clc;
figure();
a=imread('cameraman.tif');
b=double(a)*.5;
subplot(2,2,1),imshow(a),title('original image');
subplot(2,2,2),imshow(uint8(b)),title('contrast  stretched image');
```

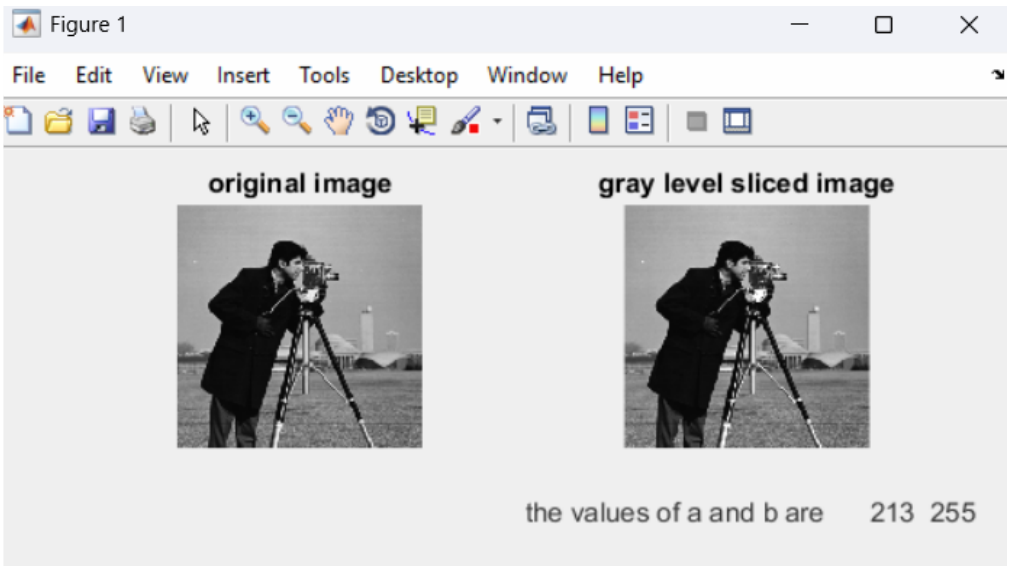
OUTPUT:



3.2 Gray level slicing

```
clear all;
close all; clc;
f=imread('cameraman.tif');
f=double(f);
[m,n]=size(f);
L=double(255);
a=double(round(L/1.2));
b=double(round(L));
for i=1:m
   for j=1:n
      if (f(i,j)>=a & f(i,j)<=b)
         g(i,j)=L;
      else
         g(i,j)=f(i,j);
      end
   end
end
subplot(2,2,1),imshow(uint8(f)),title('original image');
subplot(2,2,2),imshow(uint8(g)),title('gray level sliced image');
xlabel(sprintf(' the values of a and b are      %g  %g',a,b));
```

OUTPUT:

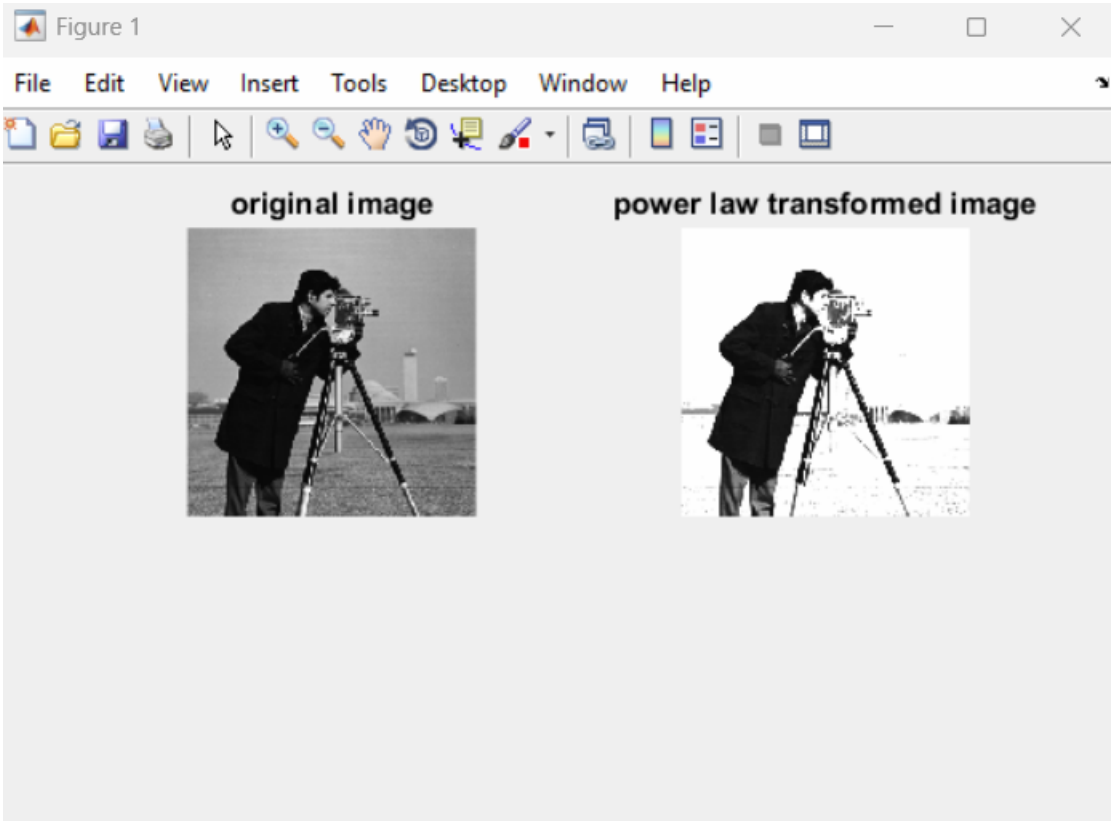


3.3 Power law transformation

```
close all;
clear all;
clc;
r=double(imread('cameraman.tif'));
g=input('enter the value of gamma = ');
s=r.^g;
subplot(221);imshow(uint8(r));title('original image');
subplot(222);imshow(uint8(s));title('power law transformed image');
```

OUTPUT:

```
enter the value of gamma  1.2
```



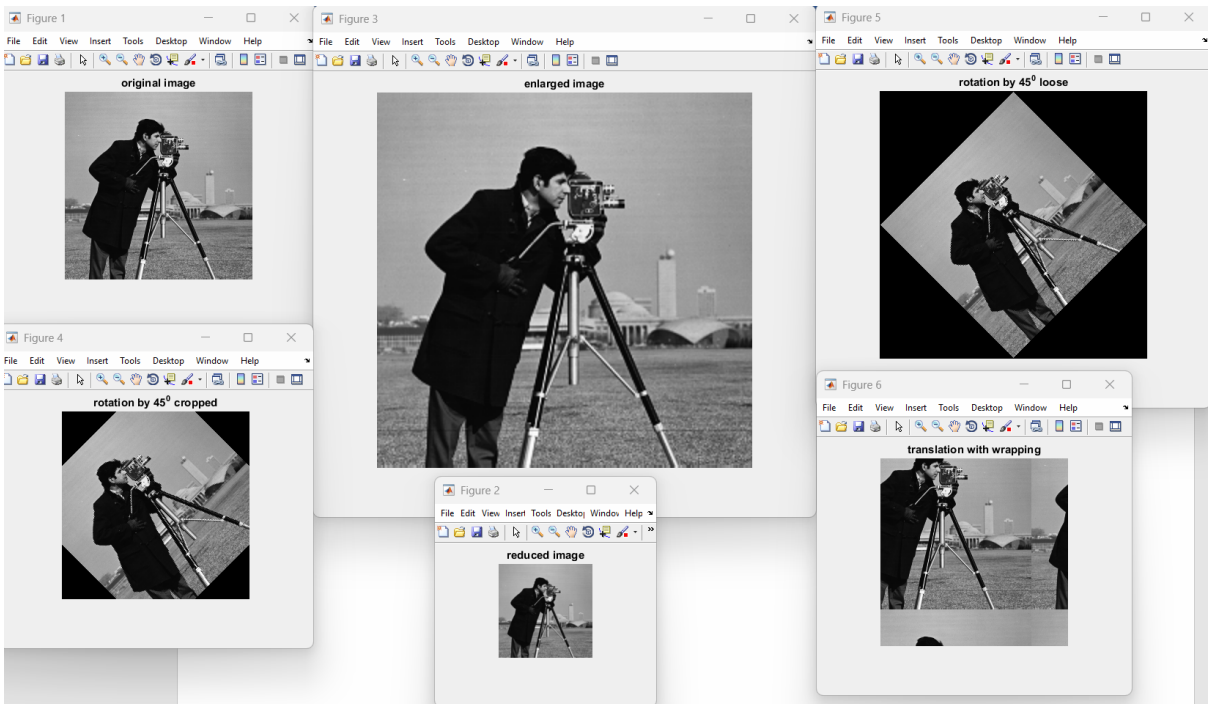
PROGRAM 4: Basic transformation of the given image such as resize, rotate and wrap the images

Code:

```
clc;
clear all;
I=imread('cameraman.tif');
I_reduced=imresize(I,0.5,'bilinear');
I_enlarged=imresize(I,2);
figure(1);
imshow(I);
title('original image');
figure(2);
imshow(I_reduced);
title('reduced image');
figure(3);
imshow(I_enlarged);
title('enlarged image');
I_rot1=imrotate(I,45,'crop');
I_rot2=imrotate(I,45,'loose');
figure(4);
imshow(I_rot1);
title('rotation by 45^{0} cropped');
figure(5);
imshow(I_rot2);
```

```
title('rotation by 45^{0} loose');
x0=50;
y0=50;
I_t=wshift('2d',I,[x0,y0]);
figure(6);
imshow(I_t);
title('translation with wrapping');
```

Output:

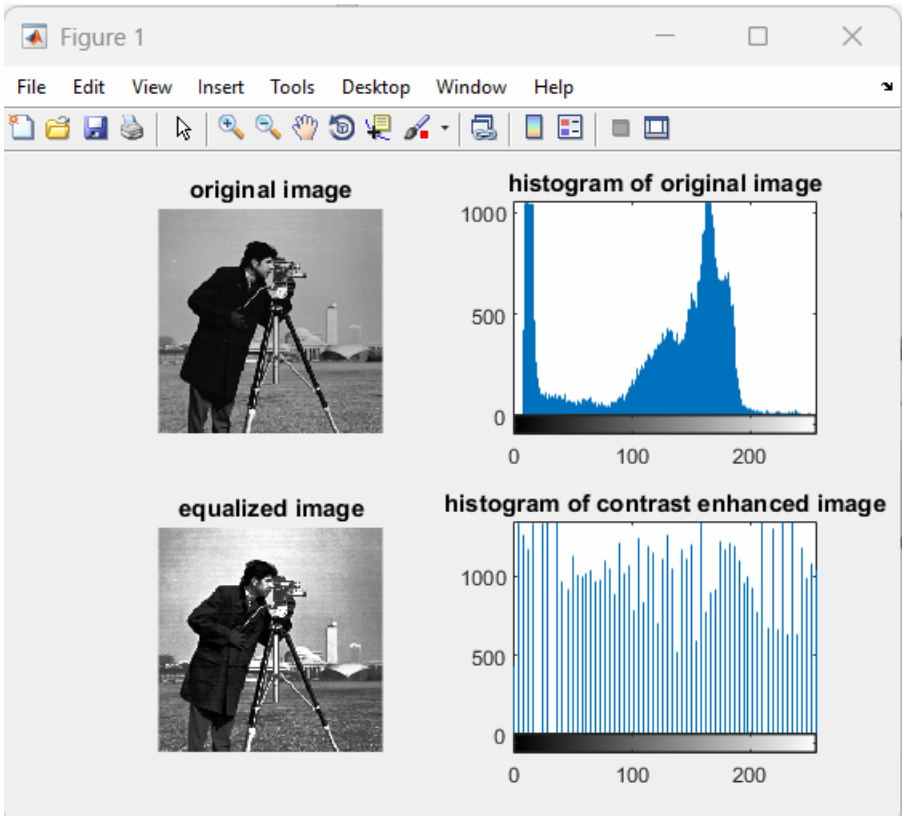


PROGRAM 5: Histogram processing and spectra in understanding the information content of images

Code:

```
clc;
clear all;
I= imread('cameraman.tif');
I2 = histeq(I);
figure;
subplot(2,2,1);
imshow(I);
title('original image');
subplot(2,2,2);
imhist(I);
title(' histogram of original image');
subplot(2,2,3);
imshow(I2);
title(' equalized image');
subplot(2,2,4);
imhist(I2);
title('histogram of contrast enhanced image');
```

Output:


Figure 1
File Edit View Insert Tools Desktop Window Help
original image
histogram of original image
1000
500
0
0 100 200
equalized image
histogram of contrast enhanced image
1000
500
0
0 100 200

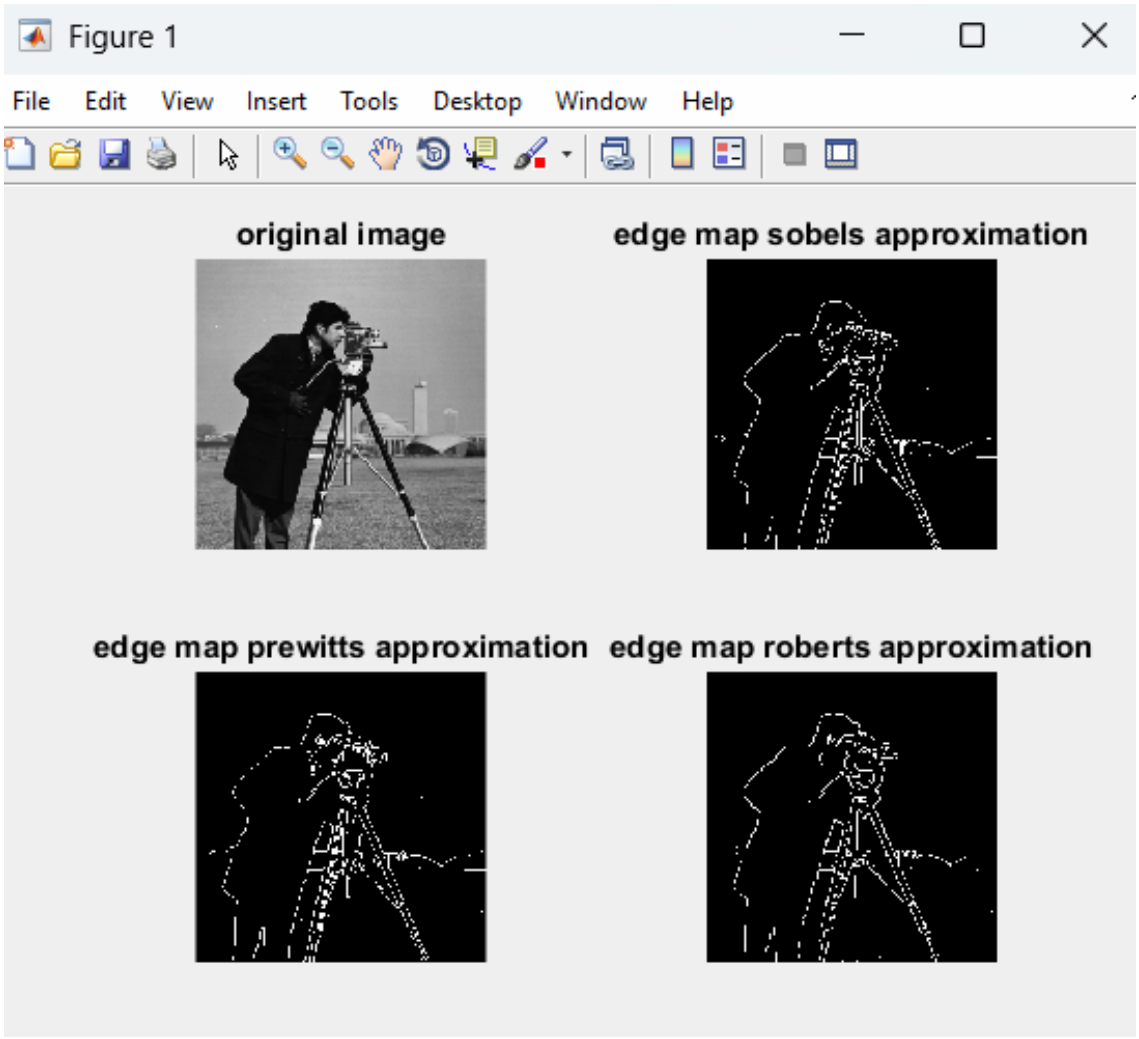
PROGRAM 6: Edge detection and filtering of an image

6.1 Edge detection of a given image using Sobel, Prewitt and Roberts edge detection techniques

Program

```
clc;
clear all;
I= imread('cameraman.tif');
Iout1= edge(I, 'sobel');
Iout2= edge(I, 'prewitt');
Iout3= edge(I, 'roberts');
figure;
subplot(2,2,1);
imshow(I);
title('original image');
subplot(2,2,2);
imshow(Iout1);
title('edge map sobels approximation');
subplot(2,2,3);
imshow(Iout2);
title('edge map prewitts approximation');
subplot(2,2,4);
imshow(Iout3);
title('edge map roberts approximation');
```

Output:

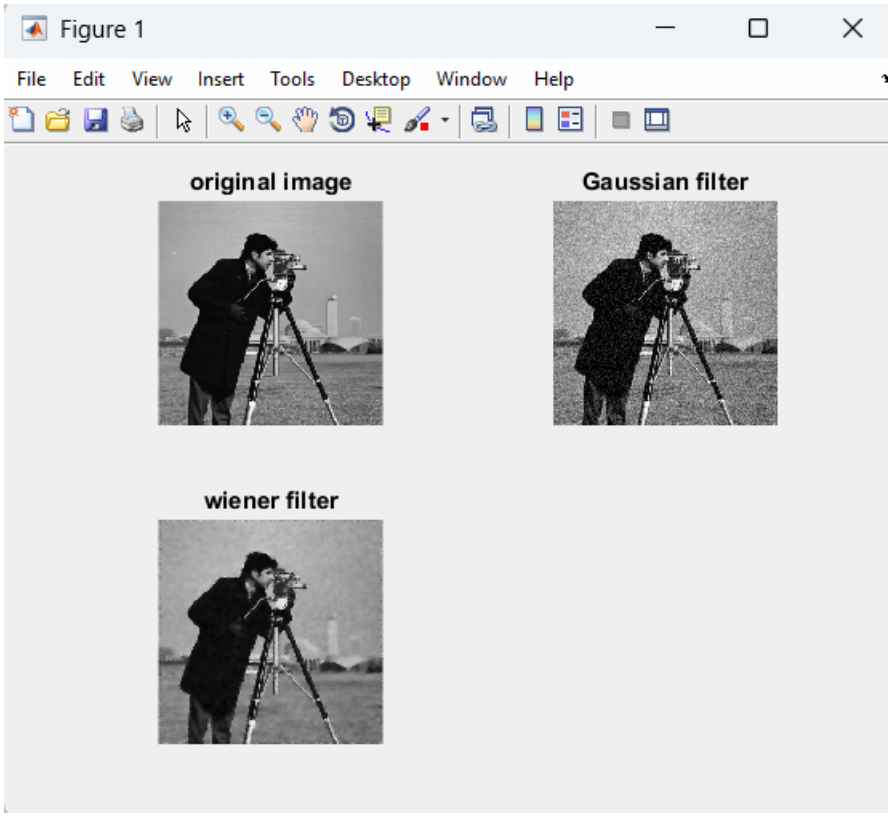

Figure 1
File Edit View Insert Tools Desktop Window Help
original image
edge map sobels approximation
edge map prewitts approximation
edge map roberts approximation

6.2 Filtering of an image using Weiner filter and Gaussian filter

Program

```
clc;
clear all;
I= imread('cameraman.tif');
J= imnoise(I, 'gaussian', 0,0.005);
K= wiener2(J,[5 5]);
figure;
subplot(2,2,1);
imshow(I);
title('original image');
subplot(2,2,2);
imshow(J);
title('Gaussian filter');
subplot(2,2,3);
imshow(K);
title('wiener filter');
```

Output:


Figure 1
File Edit View Insert Tools Desktop Window Help
original image
Gaussian filter
wiener filter